Module 3 Lecture - Claims & Validity

Research Methods in Psychology

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1 Overview

1.1 Learning Objective(s)

- Differentiate the three types of claims: frequency, association, and causal.
- Ask appropriate questions to help you interrogate each of the four big validities: construct validity, statistical validity, external validity, and internal validity.
- Explain which validities are most relevant for each of the three types of claims.
- Be able to identify types of variables, their scales of measurement, and the levels/values of a given variable. Understand what variables are likely to be manipulated vs. observed.
- Be able to identify and write frequency, association, and causal claims. Be able to assess what type of claim a certain article is attempting to make. Understand which scale of measurement variables are normally associated with each claim.
- Be able to assess what source a claim comes from, and whether the weight of evidence for a claim is valid. Understand how to approach claims through the lens of the four validities we discuss.

1.2 Lesson Overview

- Up until now, we have mostly focused on the more _____ and philosophical parts of why we conduct research
- Now, we are going to start to put more _____ names to the many moving parts of a study
- This is going to include talking about variables, claims, and validities - topics which are going to _____ over the remainder of the semester.

> **Important**
>
> Remember, psychological studies are a series of *choices* we make! Those choices should be rooted in empiricism.

2 Variables

2.1 Definition

- A variable is some _____ that can vary, with at least 2 levels/values
  - Contrast this with a constant, which does not vary at all
- Variables are important in both the _____ and the data stages of the theory-hypothesis-data cycle
  - In fact, to make _____ work, we need variables!

Discuss: Review from stats class: list off as many statistical tests as you can

- Variables (and their definitions within the context of the study) are core, critical parts of a _____ - and will play an important role in the validity of the results

2.2 Scales of Measurement

Likely a review from statistics course

- Every variable has some _____ of how it is measured:
- Interval/ratio/continuous: A variable has _____, known distances between each level/value, arranged in *known* order.
  - Example: Age, height in centimeters, point score on test.
- Ordinal A variable has _____ distances between each level/value, arranged in a *known* order.
  - Example: Place in a foot race, Class rank
- Categorical/Nominal : A variable's levels/values are entirely _____ from one another, arbitrary (no) _____ of order
  - Example: Experimental group vs. Control group; Undergraduate vs graduate students; different job titles

? I measured the income level of every family in Kent county, and then I rank the families as highest income - 1, second highest income - 2, and so on. What type of data is this?

- (●) A) Continuous
- ( ) B) Ordinal
- ( ) C) Categorical
- ( ) D) None of the above

Explanation:

Important

Correctly identifying our variables with their correct scale of measurement is an important skill, as it determines what analyses we can use them in.

2.3 Measured vs. Manipulated Variables

- In addition to Scales of Measurement, we can also _____ data as measured or manipulated
  - These following labels are *not* mutually _____ with descriptions under scale of measurement
- Measured/observed variable a measurement of something that is not _____ or *directly* caused by the experimenter (but may be the result of something an experimenter did).
  - Often, due to ethical or _____ constraints, or simply because it is simply impossible to directly change
  - Example: SAT score, hair color, class enrollment
  - This can be any of the _____ of measurement
- Manipulated/experimental variable a measurement of something that is directly _____ by an experimenter
  - Example: putting an individual in an intervention or control group.
  - This is *usually* a _____ variable

Discuss: Say I watch folks walking around on campus, and I tally the *number* of people wearing coats. Describe whether this is measured or manipulated, and what scale of measurement it it.

- Some variables *can* be _____, depending on the setup of the study.
  - E.g., I could put people in a low stress/high stress situation (_____) or I could observe their naturally high/low stress levels

Important

Just like with scale of measurement, the setup of what variables are measured vs which are manipulated have a large impact on what study designs we can use

2.4 Constructs to Operational Variables

- Construct/conceptual/latent variables Abstract variables usually representing some _____ or notion that cannot be explicitly and _____ measured
  - However, these usually *are* the concepts we actually care to draw _____ on.
  - Example: Depression, Academic achievement/aptitude
- Operational variables Concrete variables as measured through some _____ or sense
  - This is what we can directly _____ in the study
  - Example: PHQ-9, test score
- In most research, we _____ the construct to some direct measurement tool. However, the focus of our conclusions is *not* the measured variable, but rather, the _____ that it purports to measure.
  - Example: when I assign a test, do I care *only* about the plain number of items correct a student has? NO! I care about the content mastery that the items are representative of.

Discuss: I measure GPA in high school students as an operational variable; what latent variable does that represent?

- Generally, when we talk about theories, we are trying to discuss _____; and when we are talking about study-specific hypotheses, we discuss _____ variables.
- Side note: The gap between operational to latent variables is an extremely important area that deals with _____ validity, which is different from the validity we will be discussing shortly. This area of study deals with Measurement and Test Theories, as well as psychometrics (my favorite).

3 The 3 Types of Claims

3.1 Definition

- A claim, simply, is any _____ for some idea, interaction, or relationship between two or more objects.
- Claims can be derived from scientific evidence, or any of the _____ of knowledge we discussed previously
  - But remember the pitfalls of some sources of information, and prioritize _____!

Discuss: Try to list off the other (less good) sources of information we talked about in module 2

- Claims come in 3 flavors: frequency, _____, and causal

  - All three have varying _____ to be valid, which more _____ requirements moving from frequency -> association -> causal
  - Good evidence of any claim is a _____ of the scale of measurement and the design of the study

3.2 Frequency Claims

- A frequency claim is one that deals with the rate, percent, or _____ of a certain characteristic or phenomenon.
  - Example: "roughly 25% of college students report feeling stressed about an upcoming assessment"
- These claims are usually only made upon some single observed and _____ variable or measurement, and often describes a yes-no proposition.
  - Example: 20 of 50 students like the dining hall food -> 20 students say yes to liking the food, 50 said no to the same question
  - Example: 70% of professors report having burnout

> **Important**
>
> Looking for a percent sign can be one good way to detect these claims - but that isn't always accurate!

- These claims are _____ dealing with a categorical scale of measurement
  - Stats sidebar: _____ for claims of frequency are normally done with tests such χ^2 or Fisher's exact tests
- _____ represented by pie charts or bar charts

3.3 Association Claims

- An association claims is made regarding _____ or more variables, attempting to describe or quantify the relationship between them, *without* explaining which one might _____ the other
  - Hint: look for words like "correlate", "predict", "associated with", and/or "covary" to identify these claims
  - Example: I think that people who are more extroverted tend to be friendlier
- An association between two variables can either be described as *positive*, *negative*, or *non-existent/zero:*
  - Positive relationship -> as one variable increases, so does the other

  - Negative relationship -> as one variable increases, the other _____
  - Non-existent relationship -> both variables do not consistently _____
- Variables in this claim usually have to be at least _____, but are more commonly continuous in nature. In most cases, variables in this sort of study would only be _____ or measured.
  - Stats sidebar: testing for these claims is usually done with some form of _____ test (e.g., Pearson's r, Spearman's ρ, Kendall's $\tau - b$) or regression (e.g., linear - OLS, curvlinear)

> Discuss: In the 40% of patients who report having headaches, migraine symptoms increase with lower sleep

- Graphically represented by _____ (one variable on x-axis, other on y-axis)

3.4 Causal Claims

- Causal claims similar to an association claim, with the added caveat that these claims make some _____ on which of two or more variables causes another.
  - This claim requires the most _____ and substantial evidence of all three claim types. Whereas a claim of association or frequency are largely *descriptive*, a causal claim is more *inferential*.
  - Example: I think eating more carbs makes me run better in the morning

> **Important**
>
> This is where we might say the old adage: 'correlation does not equal causation'

- Similar to associative claims, a causal claim is likely to describe the movement of one variable in _____ to another, but must meet three additional criteria:
  - The two variables are correlated (valid _____ claim)
  - One variable clearly came before the other (_____ prece-

dence)
  - The relationship or change cannot be _____ by some other factor/confound (internal validity - more on this later)

- For solid causal claims, _____ are really the most (and maybe only) acceptable design (but we will cover that later as well)
- In causal claims, we are most likely to encounter observed _____ data as our outcome, and some form of _____ experimental/-manipulated variable.
  - Stats sidebar: because of the scale of measurement just described, we are usually thinking about some sort of t-test or _____, to test means between tow or more groups.

Discuss: I am measuring ant speed in relation to how much sugar they have in their system - I am measuring each ant's natural sugar level and then their walking speed directly after. I am hypothesizing that higher sugar level causes faster walking speed. Explain why this is a causal claim AND why this setup is insufficient for proving that causal claim.

3.5 Claims Are Not Always Research

- Claims can appear in personal experience, appeals to _____, and intuition; just because a claim is made, does not mean that they are based in research
  - *Example:* "I swear this happens to me every day", "This all happened because [insert reason here]"

Discuss: What type of claim are those two above?

- If a claim is made, ask yourself, "what is the _____ of the claim?" Hopefully an empirical article!

4 Assessing Claims with 4 Types of Validity

4.1 Overview

- Alongside the theory-hypothesis-data framework we discussed early in the semester - these 4 _____ will make up the backbone on how you approach critiquing scientific claims.

> **Important**
>
> I want you to think about these 4 validities *constantly* as you plan our your research proposal!

4.1.1 Construct Validity

- Construct validity is all about how well the variables under study are _____, manipulated, and measured. This is often closely related to the operational measures one chooses to capture the latent/construct variables.
- Faults in _____ validity imply a failure to properly represent the constructs under study, and will lead to a disconnect between the study and the _____ under study
  - We will discuss more about "good" measurement _____ in week 5

> Discuss: Was this quiz at the start of class a good measurement of your learning in the last module? Why or why not?

- Example: how does a doctor measure + quantify a person's 'depression'? What tool do they use for that, and what evidence do we have that it works?

4.1.2 External Validity

- External validity, also sometimes called _____, is all about how well the claims and results of a study/hypothesis apply to the broader _____ and other contexts.
  - This tends to be primarily determined by how tightly _____ a study is (more control = less generalizable to daily life) and how individuals were _____ for a study (more on this in week 7).

> **Important**
>
> A larger sample size alone does not make a study more generalizable

- Faults in external validity lead to _____ in how well a claim can be extrapolated and abstracted outside the context of a study.
- Example: Is a tightly-controlled clinical study on mice in a lab cage readily applicable to the average person in their daily life? No!

4.1.3 Statistical Validity

- Statistical validity is all about whether the _____, p-values, effect sizes, and _____ methods support the hypothesis in question.
- Helpful questions to ask:
  - Did the authors state an α level in the methods?
  - Did the paper report adequate statistical information to assess significance? (e.g., p-values, _____ intervals, effect sizes - ideally all 3)
  - Did the authors themselves actually _____ this information correctly?
  - Was the analysis _____ enough to detect an effect? (e.g., big enough sample)
- Failures in this type of validity _____ the conclusions made in the discussion, which comes from the results.
- Example: Did the authors of this study choose an analysis method appropriate to the data and variables we have?

4.1.4 Internal Validity

- Internal validity is most closely related to how well other variables are _____ for and noted in a study. Researchers may use a combination of _____ controls (which we will discuss in this class) and statistical controls (which are more relevant to PSY-350).

> Discuss: What vocabulary word did we use to describe a problem in a third variable that might complicate our explanations? Hint: starts with a c

- Failure to properly address these will result in _____ on the actual causation/association mechanism between two or more variables.
- This tends be be one of the _____ validities to fully meet, and one that many people fail to account for in intuition and personal experience.

4.2 Claim Types and Relationship with Validity

- In my opinion, the validities don't necessarily need a tailor-made application to each type of claim. The one exception is that internal validity is not really applicable to _____ claims, and only very little to association claims.

4.3 How to Prioritize Certain Validities

> **Important**
>
> Remember, there is no such thing as the 'perfect' study!

- As we have discussed before, no study can be so well-made that it satisfies all validities and answers all questions about a _____. Instead, it is our goal to balance _____ and our available resources to create a reasonably good study that provides evidence for a part of a broader question.

  - This is important to remember both as we create research (research _____) and as we read and assess existing research (research _____)!
- I propose 4 areas to consider when choosing how to prioritize:
  - Type of _____ to be made (frequency, association, causal) - see previous sections for discussions on those.
  - _____: time, resources, funding
  - _____: sampling method, interventions, _____ of populations
  - _____: how important and _____ are our findings
- But, there is no one panacea or perfect solution to balance these all - we often do research understanding the natural _____ of our work.

5 Conclusion

5.1 Recap

- In this lecture, we covered different types of claims one can make, and the different validities to consider when we make those claims
- We also discussed how to accurately describe variables, and the general concept of how operations and latent variables are connected with one anothe
- Finally, we reinforced that careful decision-making in each of these steps helps ensure we make better science and studies

Key Terms

A

association claims A description of how two or more variables are related to one another, without describing which one causes another 7

C

Categorical/Nominal A variable scale of measurement in which data are represented by qualitative labels and categories 3
Causal claims Describe a relationship between two or more variables in which one causes another 8
claim An argument for some idea, interaction, or relationship in variable(s) 6

constant A measurement or characteristic that remains the same across the entire sample or population 3
Construct validity The validity of how our constructs are measured and manipulated 10
Construct/conceptual/latent variables Variables that are abstract and represent some idea that is not directly measurable 5

E

External validity The validity of how well our results generalize or expand to the broader population 11

F

frequency claim A claim made regarding a rate or percentage of occurrences 7

I

Internal validity The validity as to whether we can be certain that changed in one variable caused change in another 12
Interval/ratio/continuous A variable scale of measurement has consistent, distances and set order 3

M

Manipulated/experimental variable A variable that is directly changed by the experimenter 4
Measured/observed variable A variable that is gathered or measured, but *not* directly changed by the researchers 4

N

Negative relationship Where one variables goes up, the other decreases 8
Non-existent relationship Where there is not any consistent behavior in change between two variables 8

O

Operational variables Variables that are directly measured, usually in order to represent some latent variable 5
Ordinal A variable scale of measurement in which data are represented as ranks 3

P

Positive relationship Where one variable goes up, another does as well 7

S

Statistical validity How the statistical analysis methods support or don't support study conclusions 11

V

variable Some measurement or characteristic that varies within a sample or population 3

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms